Echo-Leakage Suppression for Location Privacy in ISAC with Eavesdropper Location Uncertainty

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***Abstract*—This paper investigates echo-leakage suppression for location privacy in integrated sensing and communication (ISAC) systems, preventing eavesdroppers (Eves) from inferring user equipment (UE) location via echo signals while ensuring UE communication quality of service (QoS). Considering Eve's location uncertainty, we formulate the problem as a min-max fractional programming model that minimizes Eve's maximum signal-to-noise ratio (SNR) across all possible locations, subject to the UE's SNR constraint. Faced with its non-convexity and semi-infinite objective, we develop two efficient solutions targeting optimization accuracy and computational complexity, respectively. First, we propose an optimization-oriented iterative algorithm, which reformulates the semi-infinite constraint via the S-procedure, transforms the beamforming covariance matrix into a finite-dimensional linear matrix inequality (LMI), and solves it using a bisection-aided semidefinite programming (SDP) scheme with guaranteed theoretical optimality. To reduce computational complexity, a lightweight learning-based strategy is further proposed: it constructs the beamformer by combining the maximum ratio transmission (MRT) vector and an Eve-related orthogonal channel component, while restricting the artificial noise (AN) covariance matrix to rank-one. Simulations demonstrate that both methods achieve substantial leakage reduction compared to baselines, with the learning-based strategy offering significantly lower computational time than the S-procedure-based algorithm.**

***Index Terms*—Integrated sensing and communication, location privacy, artificial noise, S-procedure, learning-based one-shot optimization.**

I. Introduction

Integrated sensing and communication (ISAC) is a key enabling technology for beyond fifth-generation (B5G) and sixth-generation (6G) wireless networks [1], which supports simultaneous sensing and communication via shared radio resources and network infrastructure. Although existing works mainly optimize communication and sensing performance, pervasive ISAC deployment also brings severe location privacy threats [2]. With advanced sensing techniques, unauthorized eavesdroppers (Eves) can easily infer user equipment (UE) locations from echo signals, greatly aggravating privacy leakage risks [3]. Motivated by this, this paper aims to protect UE location privacy against Eves while ensuring communication quality of service (QoS).

Numerous studies have explored privacy protection in ISAC systems, where artificial noise (AN) is a mainstream anti-eavesdropping solution. In [4], by deriving the posterior Cramér-Rao bound (CRB) for sensing mean square error (MSE), the authors design an AN-aided beamforming framework to maximize the worst-case secrecy rate under CRB constraints, achieving global optimality via a two-stage strategy. In [5], CRB is adopted to guarantee legitimate sensing performance, and semidefinite relaxation (SDR) is used to iteratively optimize beamformers and AN covariance for minimizing Eve's received SNR. In [6], SDR combined with fractional programming is utilized to minimize Eve's signal-to-interference-plus-noise ratio (SINR) while satisfying legitimate users' SINR requirements, considering both perfect and imperfect CSI conditions. In [7], a new angular-domain peak-to-average ratio metric is defined to evaluate sensing resistance. The authors optimize communication performance under Angular-Domain Peak-to-Average Ratio(ADPAR) constraints by integrating null-space projection with AN, and derive closed-form solutions for typical scenarios. For mobile Eve scenarios, [8] formulates a non-cooperative Stackelberg game, adopts successive convex approximation (SCA) for network resource optimization, and leverages deep reinforcement learning (DRL) to predict Eve's trajectory, thereby improving communication security and sensing accuracy.

Unlike AN-based schemes, [9] proposes a collaborative transmission scheme for multiple access points (APs) to protect user location privacy, adopting the expectation–maximization algorithm for beamforming optimization and privacy leakage suppression. [10], [11] exploit interference to defend against eavesdroppers and improve sensing accuracy under communication security constraints, where [10] achieves globally optimal beamforming via a three-step design. [12] presents a feasible paradigm that uses sensing signals as AN to degrade eavesdropping channels without harming communication performance. Transformer-based time series prediction is leveraged in [13] to build a deep neural network for predictive beamforming, balancing the secrecy rate and CRB for potential Eves. [14] develops a diffusion-model-based secure sensing framework. Its discrete diffusion model properly activates ISAC links and nodes, while the continuous module generates safeguarding signals to ensure accurate sensing only for authorized users. Furthermore, intelligent reflecting surfaces are applied for privacy preservation, reconfiguring propagation environments to limit unauthorized sensing and boost communication quality [15], [16].

Notably, most existing works assume exact Eve location prior, which is impractical in real scenarios. Conventional beamforming optimization relies on iterative algorithms with heavy computational overhead. Although AI-enabled methods are adopted to reduce complexity, most still require Eve trajectory or direction prediction [8], [13], [16], bringing extra computational burden. To tackle these issues, this paper con-

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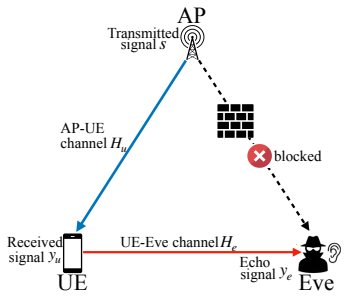


Fig. 1. ISAC privacy system model.

siders Eve's location uncertainty for echo leakage suppression in ISAC systems. We develop two beamforming paradigms: a high-precision iterative algorithm and a lightweight AI-based strategy, to balance optimization accuracy and computational efficiency. The optimization-oriented algorithm adopts the S-procedure to reformulate the semi-infinite constraint of the Eve SINR, and convert it into a finite-dimensional linear matrix inequality (LMI). It then leverages a bisection-aided SDP scheme to solve the problem with guaranteed theoretical optimality. Furthermore, a lightweight learning-based strategy is proposed to reduce computational complexity. It designs the beamformer by combining the MRT vector and Eve-related orthogonal channel component, and restricts the AN covariance matrix to rank-one. Simulations verify that both methods outperform baselines including gradient descent and zero-forcing beamformer in leakage suppression, and the learning-based strategy achieves lower computational latency than the S-procedure-based algorithm.

In what follows, Section II presents the system model and problem formulation. Sections III and IV elaborate on the S-procedure-based method and the learning-based strategy, respectively. Simulation results are provided in Section V, and Section VI draws the conclusions.

II. System Model

Consider an ISAC system with a single AP and a single UE as shown in Fig. 1. The AP is equipped with N_t transmit antennas, while the UE has N_r receive antennas. An Eve with M_e receive antennas aims to infer the UE's location information via its echo signal.

The channel between the AP and the UE is denoted as $\mathbf{H}_u \in \mathbb{C}^{N_r \times N_t}$, while the channel between the UE and the Eve is assumed to be line-of-sight (LoS), and given by

$$\mathbf{H}_e = \alpha\sqrt{\beta}\mathbf{a}_e(\psi_e)\mathbf{a}_u^H(\phi_u) \in \mathbb{C}^{M_e \times N_r}, \tag{1}$$

where α is the antenna gain, β the path-loss factor for the channel between UE and Eve, and ψ_e, ϕ_u the angle of arrival (AoA) and angle of departure (AoD) from UE to Eve, respectively. In (1), $\mathbf{a}_e(\psi_e)$ and $\mathbf{a}_u(\phi_u)$ are the receive and transmit steering vectors at Eve and UE, respectively, expressed as

$$\mathbf{a}_e(\psi_e) = \frac{1}{\sqrt{M_e}}\left[1, e^{j\frac{2\pi d_e}{\lambda}\sin\psi_e}, \ldots, e^{j\frac{2\pi d_e}{\lambda}(M_e-1)\sin\psi_e}\right]^T,$$

$$\mathbf{a}_u(\phi_u) = \frac{1}{\sqrt{N_r}}\left[1, e^{j\frac{2\pi d_u}{\lambda}\sin\phi_u}, \ldots, e^{j\frac{2\pi d_u}{\lambda}(N_r-1)\sin\phi_u}\right]^T,$$

where d_e and d_u are the adjacent antenna spacing at Eve and UE, respectively. We assume the channel between the AP and the Eve is blocked, as Eve remains hidden from the AP's LoS, as shown in Fig. 1.

Owing to the uncertainty in Eve's location and antenna gain from the perspective of the AP, we assume that $\mathbf{a}_e(\psi_e)$ and $\alpha^*\sqrt{\beta}\mathbf{a}_u(\phi_u)$ lie within spherical uncertainty sets defined as

$$\mathcal{U}_e \triangleq \left\{\mathbf{a}_e(\psi_e) : \|\mathbf{a}_e(\psi_e) - \hat{\mathbf{a}}_e\|^2 \le \eta\right\}, \tag{2a}$$

$$\mathcal{U}_u \triangleq \left\{\alpha^*\sqrt{\beta}\mathbf{a}_u(\phi_u) : \left\|\alpha^*\sqrt{\beta}\mathbf{a}_u(\phi_u) - \hat{\alpha}^*\sqrt{\hat{\beta}}\hat{\mathbf{a}}_u\right\|^2 \le \epsilon\right\}, \tag{2b}$$

respectively. Here, $\hat{\mathbf{a}}_e$ and $\hat{\alpha}^*\sqrt{\hat{\beta}}\hat{\mathbf{a}}_u$ denote the centers of $\mathcal{U}_e$ and $\mathcal{U}_u$, while η and ϵ represent the radius of the respective uncertainty balls.

Combining the channel from AP and UE and the channel from UE and Eve, the cascaded channel between AP and Eve is given by

$$\mathbf{H}_e\mathbf{H}_u = \alpha\sqrt{\beta}\mathbf{a}_e(\psi_e)\mathbf{a}_u^H(\phi_u)\mathbf{H}_u. \tag{3}$$

Combining (3) and (2b), we obtain

$$\left\|\alpha\sqrt{\beta}\mathbf{a}_u^H(\phi_u)\mathbf{H}_u - \hat{\alpha}\sqrt{\hat{\beta}}\hat{\mathbf{a}}_u^H\mathbf{H}_u\right\|^2 \le \epsilon\lambda_u, \tag{4}$$

which characterizes the feasible region of $\alpha\sqrt{\beta}\mathbf{a}_u^H(\phi_u)\mathbf{H}_u$, where λ_u denotes the maximum eigenvalue of $\mathbf{H}_u\mathbf{H}_u^H$.

During the communication phase, the UE is assumed to request a single data stream, denoted by s with $\mathbb{E}[|s|^2] = 1$, from the AP. To prevent information leakage, the AP transmits the information symbol s to the UE along with AN. The aggregated transmit signal is given by

$$\mathbf{x} = \mathbf{w}\,s + \mathbf{n}_{\text{AN}}, \tag{5}$$

where $\mathbf{w} \in \mathbb{C}^{N_t}$ is the transmit beamforming vector, and $\mathbf{n}_{\text{AN}} \sim \mathcal{CN}(\mathbf{0}, \mathbf{Z})$ is the AN following a circularly symmetric complex Gaussian distribution with covariance matrix $\mathbf{Z} \succeq 0$ to be designed. The beamforming vector and the AN should satisfy the power constraint at the AP, i.e., $\mathbf{w}^H\mathbf{w} + \text{Tr}(\mathbf{Z}) \le P_{\text{tot}}$. Based on (5) and the channel modeling, the received signals of the UE and the Eve are given by

$$\text{UE: } \mathbf{y}_u = \mathbf{H}_u(\mathbf{w}s + \mathbf{n}_{\text{AN}}) + \mathbf{n}_u, \tag{6a}$$

$$\text{Eve: } \mathbf{y}_e = \mathbf{H}_e\mathbf{H}_u(\mathbf{w}s + \mathbf{n}_{\text{AN}}) + \mathbf{n}_e, \tag{6b}$$

respectively, where $\mathbf{n}_u$ and $\mathbf{n}_e$ are the additive white Gaussian noise satisfying $\mathbf{n}_u \sim \mathcal{CN}(0, \sigma_u^2\mathbf{I})$ and $\mathbf{n}_e \sim \mathcal{CN}(0, \sigma_e^2\mathbf{I})$.

We assume that the UE employs maximum ratio combining (MRC) for message decoding, where the combiner is selected as the dominant left singular vector of $\mathbf{H}_u$, denoted by $\mathbf{v}_u$. Accordingly, we define the MRC-processed effective channel as $\hat{\mathbf{h}}_u = \mathbf{H}_u^H\mathbf{v}_u$. The resulting UE SNR is given by

$$\text{SNR}_u = \frac{|\hat{\mathbf{h}}_u^H\mathbf{w}|^2}{\hat{\mathbf{h}}_u^H\mathbf{Z}\hat{\mathbf{h}}_u + \sigma_u^2}. \tag{7}$$

On the other hand, the SNR at the Eve is given by

$$\begin{aligned}\text{SNR}_e &= \frac{\mathbf{w}^H(\mathbf{H}_e\mathbf{H}_u)^H\mathbf{H}_e\mathbf{H}_u\mathbf{w}}{\text{Tr}(\mathbf{H}_e\mathbf{H}_u\mathbf{Z}(\mathbf{H}_e\mathbf{H}_u)^H + \sigma_e^2\mathbf{I})} \\ &= \frac{\mathbf{w}^H\bar{\mathbf{a}}\mathbf{a}_e(\psi_e)^H\mathbf{a}_e(\psi_e)\bar{\mathbf{a}}^H\mathbf{w}}{\text{Tr}(\mathbf{a}_e(\psi_e)\bar{\mathbf{a}}^H\mathbf{Z}\bar{\mathbf{a}}\mathbf{a}_e(\psi_e)^H) + M_e\sigma_e^2}\end{aligned}$$

$$=\frac{|\bar{\mathbf{a}}^H\mathbf{w}|^2}{\bar{\mathbf{a}}^H\mathbf{Z}\bar{\mathbf{a}}+M_e\sigma_e^2},$$

where $\bar{\mathbf{a}} \triangleq \alpha^*\sqrt{\beta}\mathbf{H}_u^H\mathbf{a}_u(\phi_u)$ whose feasible region is given in (4).

To ensure secure transmission for the UE, our objective is to minimize the maximum achievable SNR at Eve within the potential area in (4), while guaranteeing the target received SNR at the UE. The problem is formulated as

$$\min_{\mathbf{w},\mathbf{Z}} \max_{\bar{\mathbf{a}}\in\mathcal{U}_a} \frac{|\bar{\mathbf{a}}^H\mathbf{w}|^2}{\bar{\mathbf{a}}^H\mathbf{Z}\bar{\mathbf{a}}+M_e\sigma_e^2} \tag{8a}$$

$$\text{s.t. } \frac{|\hat{\mathbf{h}}_u^H\mathbf{w}|^2}{\hat{\mathbf{h}}_u^H\mathbf{Z}\hat{\mathbf{h}}_u+\sigma_u^2} \geq \gamma_{\text{th}}, \tag{8b}$$

$$\|\mathbf{w}\|^2 + \text{Tr}(\mathbf{Z}) \leq P_{\text{tot}}, \tag{8c}$$

$$\mathbf{Z} \succeq 0, \tag{8d}$$

where γ_{th} denotes the target SNR threshold for the UE and

$$\mathcal{U}_a \triangleq \left\{ \bar{\mathbf{a}} : \left\| \bar{\mathbf{a}} - \hat{\alpha}\sqrt{\hat{\beta}}\hat{\mathbf{a}}_u^H\mathbf{H}_u \right\|^2 \leq \epsilon\lambda_u \right\}. \tag{9}$$

Problem (8) is a non-convex fractional programming problem with composite-vector uncertainty, which is difficult to solve in terms of both optimization accuracy and computational complexity. To tackle this challenge, this paper proposes two approaches to address the above issues. The next section develops a performance-oriented iterative algorithm: it losslessly reformulates the semi-infinite constraint via the S-procedure, and converts the beamforming covariance matrix into an SDP feasibility problem, whose tightness is theoretically proved. To reduce complexity further, Section IV devises a lightweight learning-based strategy by simplifying beamformer and AN covariance matrix structures.

III. S-Procedure-Based Optimal Solution

The key challenge for Problem (8) is the inner maximization over $\bar{\mathbf{a}} \in \mathcal{U}_a$ in (8a), which involves an infinite set of quadratic terms. To tackle this, we first use the S-procedure to convert the semi-infinite problem into a finite-dimensional LMI, and then solve the reformulated problem with a bisection-aided SDP method, which yields an optimal solution up to the prescribed tolerance.

A. S-Procedure

First, by introducing an epigraph variable $t \geq 0$, the inner maximization in (8) is equivalent to

$$\min\ t, \quad \text{s.t. } |\bar{\mathbf{a}}^H\mathbf{w}|^2 \leq t\left(\bar{\mathbf{a}}^H\mathbf{Z}\bar{\mathbf{a}}+M_e\sigma_e^2\right),\ \forall\,\bar{\mathbf{a}}\in\mathcal{U}_a. \tag{10}$$

By defining two quadratic functions for $\bar{\mathbf{a}}$:

$$f_1(\bar{\mathbf{a}}) \triangleq \epsilon^2\lambda_u^2 - \|\bar{\mathbf{a}}-\hat{\bar{\mathbf{a}}}\|^2 \geq 0, \tag{11a}$$

$$f_2(\bar{\mathbf{a}}) \triangleq t\left(\bar{\mathbf{a}}^H\mathbf{Z}\bar{\mathbf{a}}+M_e\sigma_e^2\right) - \bar{\mathbf{a}}^H\mathbf{w}\mathbf{w}^H\bar{\mathbf{a}} \geq 0, \tag{11b}$$

where $\hat{\bar{\mathbf{a}}} \triangleq \hat{\alpha}^*\sqrt{\hat{\beta}}\mathbf{H}_u^H\hat{\mathbf{a}}_u$, constraint (10) is equivalent to the implication "$f_1 \geq 0 \implies f_2 \geq 0$". Given $f_1(\hat{\bar{\mathbf{a}}}) = \epsilon^2\lambda_u^2 > 0$, the S-procedure is applicable, i.e., this implication holds if and only if there exists a scalar $\lambda \geq 0$ such that $f_2(\bar{\mathbf{a}}) - \lambda f_1(\bar{\mathbf{a}}) \geq 0$ for all $\bar{\mathbf{a}}$. By defining the augmented vector $\tilde{\mathbf{a}} = [\bar{\mathbf{a}}^T, 1]^T$, this constraint reduces to the following LMI

$$\mathbf{F} \triangleq \begin{bmatrix} t\mathbf{Z}-\mathbf{w}\mathbf{w}^H+\lambda\mathbf{I} & -\lambda\hat{\bar{\mathbf{a}}} \\ -\lambda\hat{\bar{\mathbf{a}}}^H & tM_e\sigma_e^2+\lambda(\|\hat{\bar{\mathbf{a}}}\|^2-\epsilon^2) \end{bmatrix} \succeq \mathbf{0}. \tag{12}$$

By converting the inner maximization in (8) into the LMI constraint (12), our objective becomes finding the minimal $t \geq 0$ such that constraints (8b)–(8d) and (12) are satisfied.

B. Bisection SDP

In this subsection, We employ a bisection method to minimize t subject to constraints (8b)–(8d) and (12). For any fixed t, we first introduce a rank-one matrix $\mathbf{W} = \mathbf{w}\mathbf{w}^H$ and formulate the following feasibility SDP problem:

$$\text{find } (\mathbf{W}, \mathbf{Z}, \lambda)$$

$$\text{s.t. } \begin{bmatrix} t\mathbf{Z}-\mathbf{W}+\lambda\mathbf{I} & -\lambda\hat{\bar{\mathbf{a}}} \\ -\lambda\hat{\bar{\mathbf{a}}}^H & tM_e\sigma_e^2+\lambda(\|\hat{\bar{\mathbf{a}}}\|^2-\epsilon^2) \end{bmatrix} \succeq \mathbf{0}, \tag{13a}$$

$$\hat{\mathbf{h}}_u^H\mathbf{W}\hat{\mathbf{h}}_u \geq \gamma_{\text{th}}\left(\hat{\mathbf{h}}_u^H\mathbf{Z}\hat{\mathbf{h}}_u+\sigma_u^2\right), \tag{13b}$$

$$\text{Tr}(\mathbf{W}) + \text{Tr}(\mathbf{Z}) \leq P_{\text{tot}}, \tag{13c}$$

$$\mathbf{W} \succeq \mathbf{0},\ \mathbf{Z} \succeq \mathbf{0},\ \lambda \geq \mathbf{0}. \tag{13d}$$

by ignoring the rank-one constraint on $\mathbf{W}$.

Problem (13) is a convex feasibility problem which can be solved via standard convex optimization solvers such as CVX. Though the rank-one constraint on $\mathbf{W}$ is relaxed therein, the subsequent proposition proves that for any feasible solution $(\mathbf{W}, \mathbf{Z}, \lambda)$ to (13), a beamforming vector $\mathbf{w}$ always exists to keep $(\mathbf{w}\mathbf{w}^H, \mathbf{Z}, \lambda)$ feasible.

Proposition 1. *For any given t, if $(\mathbf{W}^\star, \mathbf{Z}^\star, \lambda^\star)$ is feasible for Problem* (13)*, then $(\mathbf{w}^\star(\mathbf{w}^\star)^H, \mathbf{Z}^\star, \lambda^\star)$ remains feasible with*

$$\mathbf{w}^\star = \mathbf{W}^\star\hat{\mathbf{h}}_u \big/ \sqrt{\hat{\mathbf{h}}_u^H\mathbf{W}^\star\hat{\mathbf{h}}_u}. \tag{14}$$

Proof. We first define $\mathbf{p} = (\mathbf{W}^\star)^{1/2}\hat{\mathbf{h}}_u \big/ \sqrt{\hat{\mathbf{h}}_u^H\mathbf{W}^\star\hat{\mathbf{h}}_u}$ which satisfies $\|\mathbf{p}\| = 1$ and $\mathbf{w}^\star(\mathbf{w}^\star)^H = (\mathbf{W}^\star)^{1/2}\mathbf{p}\mathbf{p}^H(\mathbf{W}^\star)^{1/2}$. We then obtain

$$\hat{\mathbf{h}}_u^H\mathbf{w}^\star(\mathbf{w}^\star)^H\hat{\mathbf{h}}_u = \hat{\mathbf{h}}_u^H\mathbf{W}^\star\hat{\mathbf{h}}_u,$$

$$\mathbf{W}^\star - \mathbf{w}^\star(\mathbf{w}^\star)^H = (\mathbf{W}^\star)^{1/2}(\mathbf{I}-\mathbf{p}\mathbf{p}^H)(\mathbf{W}^\star)^{1/2} \succeq \mathbf{0},$$

which implies that constraints (13b) and (13c) are satisfied. Furthermore, it holds that

$$\mathbf{F}\big(\mathbf{w}^\star(\mathbf{w}^\star)^H, \mathbf{Z}, \lambda\big) = \mathbf{F}\big(\mathbf{W}^\star, \mathbf{Z}, \lambda\big) + \text{diag}\big(\mathbf{W}^\star - \mathbf{w}^\star(\mathbf{w}^\star)^H, 0\big) \succeq 0$$

which ensures that constraint (13a) is satisfied. This completes the proof of Proposition 1. □

Based on the theoretical guarantee of Proposition 1, Algorithm 1 outlines the proposed scheme. We conduct a bisection search for t over the interval $[0, t_{\text{ub}}]$, and solve Problem (13) for each candidate t to find the global optimum $t^\star$ of Problem (8) within tolerance δ. The optimal beamformer $\mathbf{w}^\star$ is then derived in closed form via Proposition 1. The upper bound t_{ub} in Algorithm 1 is derived from the following inequality:

$$t^* = \max \frac{|\bar{\mathbf{a}}^H\mathbf{w}|^2}{\bar{\mathbf{a}}^H\mathbf{Z}\bar{\mathbf{a}}+M_e\sigma_e^2} \leq \frac{(\|\hat{\bar{\mathbf{a}}}\|+|\epsilon\lambda_u|)^2 P_{\text{tot}}}{M_e\sigma_e^2}.$$

according to (9) and (8c).

Algorithm 1 Bisection-Based Optimal Solution

Input: $\hat{\mathbf{h}}_u$, $\hat{\bar{\mathbf{a}}}$, ϵ, $\gamma_{\rm th}$, $P_{\rm tot}$, σ_u^2, σ_e^2, tolerance δ
Output: $\mathbf{w}^\star$, $\mathbf{Z}^\star$, $t^\star$

Set $t_{\rm lb}\leftarrow 0$, $t_{\rm ub}\leftarrow (\|\hat{\bar{\mathbf{a}}}\|+|\epsilon\lambda_u|)^2 P_{\rm tot}/(M_e\sigma_e^2)$
Solve the fixed-level feasibility SDP (13) at $t_{\rm ub}$
if infeasible **then return** outage, **else**
Store the feasible initialization $(\mathbf{W}^\star, \mathbf{Z}^\star, \lambda^\star)$
while $t_{\rm ub}-t_{\rm lb}>\delta$ **do**
  $t\leftarrow (t_{\rm lb}+t_{\rm ub})/2$
  Solve the fixed-level feasibility SDP (13) at level t
  if feasible **then**
    $t_{\rm ub}\leftarrow t$, and store the feasible SDP solution
  else
    $t_{\rm lb}\leftarrow t$
  end if
end while
Recover $\mathbf{w}^\star \leftarrow \mathbf{W}^\star\hat{\mathbf{h}}_u/\sqrt{\hat{\mathbf{h}}_u^H\mathbf{W}^\star\hat{\mathbf{h}}_u}$
return $\mathbf{w}^\star$, $\mathbf{Z}^\star$, $t^\star\leftarrow t_{\rm ub}$

Remark 1. *(Complexity) For each fixed t, Algorithm 1 solves one feasibility SDP with two $N_t\times N_t$ Hermitian matrix variables and one scalar variable. Since the problem contains one (N_t+1)-dimensional LMI together with the semidefinite constraints on $\mathbf{W}$ and $\mathbf{Z}$, a standard interior-point implementation requires $\mathcal{O}(N_t^{6.5})$ complexity per fixed-t solve. Hence the overall complexity is $\mathcal{O}\big(N_t^{6.5}\log_2((t_{\rm ub}-t_{\rm lb})/\delta)\big)$.*

IV. Learning-based One-shot Solution

Although Algorithm 1 acts as an optimal offline oracle, its iterative optimization leads to high computational complexity. To address this issue, we propose a learning-based one-shot design rule to construct beamformer $\mathbf{w}$ and AN covariance matrix $\mathbf{Z}$. Notably, this scheme no longer involves the oracle-level variable t, and can effectively reduce computational overhead by simplifying the structures of $\mathbf{w}$ and $\mathbf{Z}$.

A. Variable Reduction

We focus on the two key vectors $\hat{\mathbf{h}}_u$ and $\hat{\bar{\mathbf{a}}}$ in Problem (8) for designing the beamforming vector $\mathbf{w}$ and the AN covariance matrix $\mathbf{Z}$. Recall that $\hat{\mathbf{h}}_u$ denotes the effective channel between the UE and the AP after MRC processing, while $\hat{\bar{\mathbf{a}}}$ is the center of the feasible region $\mathcal{U}_a$ for $\bar{\mathbf{a}}$.

From the perspective of beamforming design, it is intuitive from (8a) and (8b) that $\mathbf{w}$ should be selected to maximize its inner product with $\hat{\mathbf{h}}_u$ while minimizing its inner product with $\hat{\bar{\mathbf{a}}}$. To accommodate these two conflicting design criteria for $\mathbf{w}$, we introduce two auxiliary vectors: the MRT vector $\mathbf{u}\triangleq\hat{\mathbf{h}}_u/\|\hat{\mathbf{h}}_u\|$ and the orthogonal component $\mathbf{a}_\perp\triangleq(\mathbf{I}-\mathbf{u}\mathbf{u}^H)\hat{\bar{\mathbf{a}}}$. Here, $\mathbf{a}_\perp$ captures the pure Eve-related channel component in $\bar{\mathbf{a}}$ by projecting out the component aligned with $\mathbf{u}$. We then construct the beamforming vector $\mathbf{w}$ in the form

$$\mathbf{w}=\sqrt{p_w}\,\frac{\mathbf{u}-e^{-j\angle(\mathbf{u}^H\hat{\bar{\mathbf{a}}})}\kappa\mathbf{v}}{\big\|\mathbf{u}-e^{-j\angle(\mathbf{u}^H\hat{\bar{\mathbf{a}}})}\kappa\mathbf{v}\big\|},\tag{15}$$

to balance the received signal power at the UE and the Eve, where $\mathbf{v}=\mathbf{a}_\perp/\|\mathbf{a}_\perp\|$, $p_w>0$ denotes the transmit power variable, and $\kappa>0$ denotes the weighting parameter between $\mathbf{u}$ and $\mathbf{v}$. Since we have $|\hat{\bar{\mathbf{a}}}^H\hat{\mathbf{w}}|=\big|\hat{\bar{\mathbf{a}}}^H\mathbf{u}-\kappa\|\mathbf{a}_\perp\|\big|$, the term $e^{-j\angle(\mathbf{u}^H\hat{\bar{\mathbf{a}}})}\kappa\mathbf{v}$ in the numerator of (15) is intended to suppress the received power at Eve caused by $\mathbf{u}$, while $\mathbf{u}$ itself is designed to enhance the received signal power at the UE.

To enhance the AN power received by Eve, we restrict the AN covariance matrix to the rank-one structure $\mathbf{Z}=p_z\mathbf{v}\mathbf{v}^H$, where p_z denotes the AN transmit power. To meet the transmit power constraint, p_w and p_z should satisfy $p_w+p_z\le P_{\rm tot}$.

By adopting the rank-one structure for $\mathbf{Z}$ and the affine parameterization for $\mathbf{w}$, the number of optimization variables is reduced to three, i.e., $\{\kappa, p_w, p_z\}$, which significantly lowers the computational complexity. In the following subsection, we first adopt a heuristic approach to determine the power variables p_w and p_z, and then introduce a dedicated prediction framework for κ to further reduce the computational cost.

B. Parameter Design

We heuristically set p_w and p_z by imposing equality on constraints (8b) and (8c). This is motivated by fully utilizing the available transmit power and minimizing signal leakage from the UE to the Eve. Accordingly, these variables can be readily computed as

$$\begin{cases} p_w=\dfrac{\gamma_{\rm th}(P_{\rm tot}|\hat{\mathbf{h}}_u^H\mathbf{v}|^2+\sigma_u^2)}{|\hat{\mathbf{h}}_u^H\hat{\mathbf{w}}|^2+\gamma_{\rm th}|\hat{\mathbf{h}}_u^H\mathbf{v}|^2},\ p_z=P_{\rm tot}-p_w, & \text{if } p_w\le P_{\rm tot},\\ p_w=P_{\rm tot},\quad p_z=0, & \text{otherwise.}\end{cases}$$

For the weighting parameter κ, we use supervised learning to learn its mapping from the channel pair $(\hat{\mathbf{h}}_u,\hat{\bar{\mathbf{a}}})$. We evaluate the beamforming vector $\mathbf{w}$ in (15) over a κ grid and select the optimal $\kappa^\star$ that minimizes worst-case signal leakage under multiple perturbations sampled from $\mathcal{U}_a$. With training data pairs $(\hat{\mathbf{h}}_u,\hat{\bar{\mathbf{a}}},\kappa^\star)$ constructed, we train a fully connected neural network (FCN) to learn the target mapping. To fully exploit wireless channel features, the network input is designed as:

$$\Big[\|\hat{\mathbf{h}}_u\|, |\mathbf{u}^H\hat{\bar{\mathbf{a}}}|, \|\mathbf{a}_\perp\|, |\mathbf{u}^H\hat{\bar{\mathbf{a}}}|/\|\mathbf{a}_\perp\|, \|\mathbf{a}_\perp\|/|\mathbf{u}^H\hat{\bar{\mathbf{a}}}|, \|\hat{\bar{\mathbf{a}}}\|\Big]^T.$$

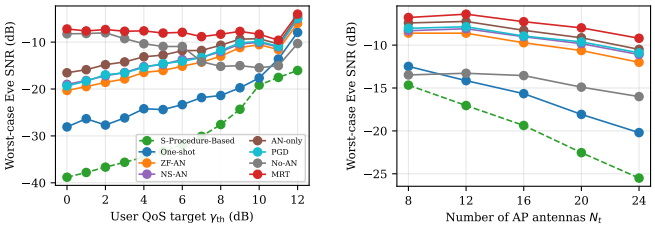
V. Simulation Results

This section evaluates the performance of the proposed methods via simulations averaged over 2000 independent scenarios. The AP, UE and Eve are uniformly distributed within a $100\,\text{m}\times 100\,\text{m}$ area with a minimum inter-node distance of 8 m. The AP–UE channel follows Rician fading with factor $K_{\rm R}=10$ dB. For the UE–Eve channel, we set $\alpha=1$ and $\beta=100/d^2$, where d is the UE–Eve distance with a 10 m reference range. Unless otherwise stated, the remaining system parameters are configured as $N_t=M_e=16$, $N_r=4$, $\lambda=1$, $d_e=d_u=0.5$, $P_{\rm tot}=1$ W, $\gamma_{\rm th}=10$ dB, $\sigma_u^2=\sigma_e^2=0.1$, and $\epsilon=0.1\|\hat{\bar{\mathbf{a}}}\|$. The one-shot method trains a 3-layer neural network with GELU activation using 5000 samples. We select κ from $\{0, 0.05, \ldots, 2\}$ for training and adopt 2000 perturbations from $\mathcal{U}_a$ to compute (8a). For the bisection-based algorithm, the tolerance is set to $\delta=5\times10^{-3}$.

We consider 6 baseline methods where all AN-based methods allocate minimum power to meet the SNR threshold in (8c) and direct the remainder to AN.

TABLE I
ACHIEVABLE SNR AND COMPUTATIONAL TIME OF DIFFERENT METHODS.

Method	Eve SNR (dB)	User SINR (dB)	Latency
S-Procedure-Based	-19.2 ± 10.5	10.0 ± 0.0	7.30 s
One-shot	-16.5 ± 10.0	10.2 ± 0.6	0.36 ms
ZF-AN	-9.7 ± 6.4	10.0 ± 0.0	16 μs
NS-AN	-9.2 ± 6.7	10.0 ± 0.0	14 μs
AN-only	-8.5 ± 6.6	10.0 ± 0.0	10 μs
PGD	-9.0 ± 6.9	10.0 ± 0.1	14.2 ms
No-AN	-14.5 ± 7.7	10.0 ± 0.0	14 μs
MRT	-7.6 ± 6.6	11.6 ± 1.2	4 μs



(a) Varying UE QoS target γ_{th}. (b) Varying AP antenna number N_t.

Fig. 2. Worst-case Eve SNR vs. UE QoS Targets & AP Antenna Numbers.

1) ZF-AN: The UE employs the MRT beamformer, and the AN beam is chosen as the vector closest to $\hat{\bar{\mathbf{a}}}$ in the null space of $\hat{\mathbf{h}}_u$ to avoid interfering with the UE;
2) NS-AN [17]: MRT beamformer for the UE, and isotropic AN in the null space of $\hat{\mathbf{h}}_u$;
3) AN-only: MRT beamformer and isotropic AN $\mathbf{Z} = (p_z/N_t)\mathbf{I}$;
4) PGD: 50-step projected gradient descent, initialized from ZF-AN;
5) No-AN: $\mathbf{w}$ in the null space of $\hat{\bar{\mathbf{a}}}$ and $\mathbf{Z}=\mathbf{0}$;
6) MRT: $\mathbf{w}=\sqrt{P_{\mathrm{tot}}}\hat{\mathbf{h}}_u/\|\hat{\mathbf{h}}_u\|$ and $\mathbf{Z}=\mathbf{0}$.

Table I compares user SNR, Eve SNR and computational latency among different schemes. All methods satisfy the required user SNR threshold. The bisection-based approach achieves the lowest Eve SNR with optimality guaranteed by Proposition 1, yet suffers high latency from iterative bisection search. The one-shot method reduces computation time by over 20,000 times, with Eve SNR rising by less than 3 dB. PGD has much higher latency and inferior Eve SNR than the one-shot scheme, showing the limited effectiveness of gradient-based optimization for the problem. Although ZF-AN, NS-AN, AN-only and MRT have extremely low runtime, their Eve SNRs are far higher than the proposed methods. The No-AN scheme obtains much lower latency than the one-shot method, with only a 2 dB Eve SNR increment due to the small number of AP antennas. However, additional simulations verify that this gap enlarges as AP antenna number grows.

Fig. 2(a) shows the worst-case Eve SNR against the UE SNR threshold γ_{th}. At low γ_{th}, the proposed S-procedure-based and one-shot methods achieve much lower Eve SNR than benchmarks, with the one-shot method outperforming the S-procedure-based scheme. This aligns with the results in Table I. As γ_{th} increases, the performance gap between our methods and benchmarks narrows, due to the shrinking feasible region of $\mathbf{w}$ and $\mathbf{Z}$.

Fig. 2(b) depicts the worst-case Eve SNR versus AP transmit antenna number N_t. For all schemes, Eve SNR declines with increasing N_t, as larger antenna arrays provide more beamforming degrees of freedom to suppress Eve's received power. At high N_t, the Eve SNR gap between the proposed methods and benchmarks widens, validating their capability to exploit the AP's spatial degrees of freedom.

VI. CONCLUSION

This paper studies echo-leakage suppression for location privacy in ISAC. We first propose an S-procedure-based iterative algorithm to convert the semi-infinite problem into a finite-dimensional LMI, and solve it via bisection-aided SDP. To reduce computational overhead, we further develop a lightweight FCN-based one-shot framework, simplifying beamforming and AN covariance structures to lower complexity. Simulations verify that both strategies outperform benchmark schemes.

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